

Appendix A Convergence Diagnostics

Summary of `appendix_diagnostics.py`

Produces the acceptance rate and burn-in convergence diagnostics for the uniform-measure sampler cited in Appendix A §A.2–A.3 of the paper.

Dependencies: `mc_uniform_fast.py`, `partition_numpy.py`. No external dependencies beyond numpy.

What it produces

(1) Acceptance rates

Runs 200,000 steps after a short equilibration at each production L and reports the fraction of accepted proposals. Expected range 0.89–0.96 (rising with L), as quoted in Appendix A §A.2.

(2) Burn-in diagnostics

Run at each L via `burnin_diagnostics(L)`, with burn-in and post-burn-in lengths both equal to $1000L$ steps and 4 independent chains:

- **Running mean:** step at which the running mean of k first enters the $\pm 1\%$ band around the second-half mean, expressed as a fraction of the burn-in length.
 - **Gelman–Rubin $\hat{R}$:** computed across 4 independent chains (all starting from the default `init_arrays` state, different seeds). Values below 1.01 pass. The paper cites $\hat{R} = 1.007$ at $L = 500$.
 - **Starting-point sensitivity:** runs the same burn-in from three starting states — under-dispersed ($k = 1$, single part), default ($k \approx 1.28\sqrt{L}$), and over-dispersed ($k = L$, all unit parts) — and reports the spread in second-half mean k . The paper cites a spread below 0.5%.
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Note on missing $L = 100$

`main()` currently calls `burnin_diagnostics` only for $L \in \{200, 500\}$. The $L = 100$ result cited in Appendix A §A.3 requires either adding `burnin_diagnostics(100)` to `main()` or running it manually. This should be fixed before the repository is finalised.